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Quenching rate constants for metastable argon, krypton, and xenon atoms by fluorine containing molecules and branching ratios for XeF* and KrF* formation

J. E. Velazco, J. H. Kolts, and D. W. Setser

Chemistry Department, Kansas State University, Manhattan, Kansas 66506
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The flowing afterglow technique was used to study the reactions of $Xe(^3P_2)$, $Kr(^3P_2)$, and $Ar(^3P_2)$ metastable atoms with small fluorine containing molecules. Fluorides from Groups III through VIII (XeF_2) were examined. Although all the fluorides have large quenching rate constants, only F_2, XeF_2, some interhalogen fluorides, and small molecules with the OF or NF bond have high branching ratios for XeF* or KrF* excimer formation. The branching ratio measurements were made via comparison to the XeCl* and KrCl* emission intensities from reaction of $Xe(^3P_2)$ and $Kr(^3P_2)$ with Cl_2, which were adopted as reference reactions. Within experimental error, the branching ratios are unity for $Kr(^3P_2)$ and $Xe(^3P_2)$ with Cl_2, F_2, and OF_2. Increasing the argon pressure from 1–40 torr gives extensive vibrational relaxation but no electronic quenching of the lowest energy excimer state of XeF*, XeCl*, KrCl*, or KrF*. Increasing pressure also reduces the intensity of the secondary emission system of KrCl* and XeCl* which implies collisional transfer from the $(^2\Pi_{3/2})$ to the $^2\Sigma^+$ excimer state. The occurrence of vibrational relaxation suggests a spontaneous radiative lifetime of ~ 50 nsec. A higher energy excimer state $(^2\Pi_{1/2})$ was identified for XeF*, XeCl*, XeBr*, XeI*, KrF*, and KrCl*. Discussion is presented regarding the chemical dynamics and chemiluminescence of these reaction systems.

I. INTRODUCTION

Since our initial report of the xenon–halide diatomic excimer emissions[1] from the two body reaction between metastable $Xe(^3P_2)$ atoms and halogen containing molecules, interest in these and similar systems has grown extremely rapidly. In independent work, Golde and Thrush[2] reported the vacuum ultraviolet emission associated with ArCl*, ArBr*, and ArO*. Subsequently this[3,4] and other laboratories[5–7] have identified all of the other Ar, Kr, and Xe–halogen excimer emissions except for ArI* and KrI*, which may be predissociated. Golde[5] has given a general discussion of these bound-free excimer emissions and tabulated some spectroscopic parameters based upon his interpretations. Conventional spectroscopic measurements from high pressure spectra have established some of the constants for the upper and lower states of XeCl and XeF.[8] Emission and absorption spectra of XeF, KrF, and XeCl have been observed using the matrix isolation technique.[9] In addition to the intrinsic interest in these new examples of excited state chemistry, these excimer transitions provide excellent possibilities for efficient high power lasers. Using *e*-beam pumping of rare gas–halogen mixtures at total pressures ranging from 0.5 to several atmospheres, laser emission has been reported for $XeBr^*(Xe + Br_2)$[10] XeF^* (from $Xe + F_2$, $Xe + NF_3$),[11,12] $XeCl^*$ $(Xe + Cl_2)$,[8a,11,12] $KrF^*(Kr + NF_3$, $Kr + F_2)$,[11–13] $ArF^*(Ar + F_2)$[13] and $KrCl^*(Kr + Cl_2)$[14] with estimated efficiencies of $\leq 15\%$.[12] Laser activity also has been observed for these systems using electric discharge excitation.[15]

An understanding of the mechanisms responsible for emissions in the laser systems requires the knowledge of rate constants and branching ratios for the relevant steps; therefore, our laboratory has undertaken the task of systematically measuring the total quenching rate constants and branching ratios for the reaction of $Ar(^3P_2)$, $Kr(^3P_2)$, and $Xe(^3P_2)$ with a variety of halogen donors. In order to meet criteria for a satisfactory laser, certain properties of the halogen donor RX are desirable: (i) The reaction $M^* + RX \rightarrow MX^* + R$ must be efficient, i.e., the branching ratio for formation of MX^* must be large. (ii) There should be little absorption by RX at the laser frequency. (iii) RX should be as nontoxic and noncorrosive as possible.

The metastable rare gas atoms react directly[1–4] with halogen-containing molecules to form the excimers, although other reactions also may contribute to the formation processes for conditions used for laser excitation. The mechanism for direct excimer formation most likely involves charge transfer with formation of M^+RX^- followed by dissociation giving $MX^* + R$. Based upon this mechanism, two obvious[4] criteria in the selection of the halogen donor are that RX should have a large electron affinity and that $D_0^0(RX) \lesssim D_0^0(MX^*) \cong 100$ kcal mole^{-1}. All molecules readily quench the Xe, Kr, and Ar metastable states[16–18]; the mechanisms of interaction are varied, and efficient formation of MX^* can be expected only for a small group of halogen donors.[4] The present work shows only F_2, some simple interhalogens, and molecules with NF or OF bonds to be good fluorine donors. With interhalogen compounds excimers of both halogens are frequently observed. These results parallel previous findings from $Ar(^3P_{0,2}) + RCl$ systems.[4] In addition to identifying good fluorine donor molecules, the present work contributes further data that should eventually facilitate the understanding of the interaction of electronically excited rare gas atoms with reagents of large electron affinity.

This paper reports rate constants and branching ratios for the reactions of fluorine-containing molecules with $Xe(^3P_2)$ and $Kr(^3P_2)$, and rate constants for the $Ar(^3P_2)$ reactions with fluorine and other halogen-containing molecules. Analysis of the excimer emission spectra permits some conclusions to be made re-

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garding the excimer excited states. The $Xe(^3P_2)$ $+XeF_2$ reaction gives a strong XeF^* emission at 260 nm, which previously was observed by Brau and Ewing.[7] The upper state is assigned as the $^2\Pi_{1/2}$ state. The corresponding state of $XeCl^*$, $XeBr^*$, XeI^*, $KrCl^*$, and KrF^* also is identified. The organization of the paper is as follows. Section IIA describes the experimental method for determining the total quenching rate constants, while Sec. IIB describes the emission measurements used to determine the branching ratios. Section IIIA summarizes the rate constants for the quenching of Ar, Kr, and Xe metastable atoms by halogen-containing molecules; the results of the branching ratios are summarized in Secs. IIIB and C. Finally in Sec. IV all these results are discussed. Of particular interest is the observation of vibrational relaxation at pressures above ~2 torr, which suggests that the radiative lifetimes of XeF^*, KrF^*, $XeCl^*$, and $KrCl^*$ in the higher vibrational levels are longer than ~50 nsec.

II. EXPERIMENTAL

A. Total quenching rate constants

The flow reactor (Pyrex glass tube with attached windows) and the flowing afterglow technique used in the determination of the total quenching rate constants are the same as previously employed.[16,17] Metastable Ar atoms are produced by passing a flow (0.152 moles/min) of Ar, purified by two low pressure molecular sieve traps maintained at 77 °K, through a hollow cathode discharge (250 V, ~1 mA) located approximately 15 cm upstream from the reagent inlet. Linear flow velocities of the order of 8000 cm/sec in a 3.1 cm id tubular reactor were achieved by a combined Roots blower and a 1000 liter/min vacuum pump. The experiment consists of monitoring (by atomic absorption spectroscopy) the metastable atom concentration along the tubular reactor for different reagent concentrations. Light from a Pen-Ray lamp (Oriel Corp.) crosses the reactor perpendicular to the gas flow, and after being mechanically chopped at 600 Hz (PAR 125) enters a 0.3 m McPherson monochromator which selects the Ar, Kr, or Xe line of interest (Ar, 811.5 nm; Kr, 811.2 nm; Xe, 881.9 nm). The atomic lines are detected by a RCA 7102 photomultiplier tube cooled to 240 °K (Products for Research), which reduced the dark current by a factor of 10^4 relative to room temperature. The output from the PM tube is amplified by a lock-in amplifier (PAR HR-8) and finally recorded on a strip chart.

The Kr and Xe metastables are produced by adding small amounts of Kr or Xe to the main Ar flow. The following technique was used to optimize the production of Kr^* and Xe^*, but still conserve Xe or Kr. With the Ar lamp in place, Kr (or Xe) (Matheson Gas Products: research grade without further purification) was added to the Ar flow, before the hollow cathode discharge, until $Ar(^3P_2)$ was no longer detectable by absorption at the first observation point. Since the $Ar(^3P_0)$ constitutes ~13% of the total Ar metastable concentration,[16] and since $Ar(^3P_0)$ reacts with Kr or Xe faster[16] than $Ar(^3P_2)$, removal of $Ar(^3P_2)$ metastables ensures complete removal of the $Ar(^3P_0)$. At this point the Ar lamp was replaced by a Kr (or Xe) lamp, and the Kr (or Xe) flow was increased until no further increase in the absorption of the 811.2 nm Kr line (881.9 nm for Xe) could be seen. Flows of ~0.6 mmol/min of Kr, as compared to only 0.2 mmol/min of Xe, were required to completely remove all of the Ar metastable atoms. This difference reflects the larger rate constant for quenching of Ar^* by Xe.[16]

Absorption of the Ar 811.5 nm line was 20% for a single traversal of light through the reactor (3.1 cm), whereas with Kr or Xe the absorption was only ~8%. In order to increase absorption for the latter, mirrors were used to obtain triple pass of light through the flow tube. Using the procedure discussed in the following section, the metastable concentrations in the rate constant apparatus were $\sim 5\times10^{10}$ atoms cm^{-3} for Ar^* and Xe^* and $\sim 2\times10^{10}$ for Kr^*.

The reagent flows were regulated by a stainless steel fine metering valve (Nupro SS-4SG) and measured by monitoring the pressure increase in a calibrated volume (576 ml) using a calibrated pressure transducer (Celesco Industries PD7). Typical reagent flow rates ranged from 1 to 20 μmoles/min. Reagent gases were either distilled or sublimed under vacuum several times before being mixed with argon. The mole fraction of reagent was typically between 0.10 and 0.15; most mixtures were stored in glass reservoirs at a total pressure of 300 torr. Fluorine and other reagents that react with Pyrex glass were stored in a stainless steel cylinder which was connected to the calibrated volume (see below) and to the shower-head mixing inlet of the reactor by $\frac{1}{4}$ in. ss tubing. Although the flow reactor is made of Pyrex, the depletion of F_2 or other fluorine-containing reagents from diffusion to and reaction with the wall is minor at the flow speeds used in this work. This effect, however, would tend to make the measurements lower limits to the true quenching rate constants.

Reagent purity is a serious problem for the corrosive chemicals. Extensive gas handling may result in the introduction of as many impurities as are removed. Although manufacturer's specifications usually are quoted to >95% purity, these must be regarded as suspect. Experiments with F_2 were done with a mixture prepared by ourselves from pure F_2 and tank Ar and from a prepared 10% mixture (Matheson). The rate constants quoted for the very reactive fluorine-containing molecules are less reliable than those normally reported from this apparatus (±10% reproducibility).

Several of the fluorine donors used in this study are extremely hazardous (with regard to toxicity and as oxidizing agents) and must be handled with great care. In the present work these reagents, after being mixed with the argon, were flowed directly into the pumps on the assumption that reaction with hot pump oil was the best means for disposal. The pumps also were vented to the hoods. Small "explosions" in the pumps were noted when evacutating the OF_2/Ar mixture from the storage reservoir.

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B. Measurement of relative emission intensities

The method is based upon comparing the total integrated excimer emission intensities from a given reaction to the intensities from a reference reaction. The references are the reactions of $Xe(^3P_2)$ and $Kr(^3P_2)$ with Cl_2 to give $XeCl^*$ and $KrCl^*$, respectively. The emission intensity ratios were related (see below) to the total quenching rate constants to obtain branching ratios. The apparatus consisted of a grease-free glass section for handling and storage of noncorrosive gases, and a stainless section for handling and storage of fluorine-containing gases. The stainless section contained the pressure transducer (Celesco Industries PD7), which serves the dual purpose of measuring flows and monitoring pressure in the reaction zone or gas handling section. Metal to glass connections consist of an inner ss standard taper joint machined in our shops and a glass outer standard taper. All metal sections were passivated by filling with pure fluorine at 500 torr for ~60 h. Passivation of the reservoirs is especially important and, as an addition measure, they were heated to 150 °C for 12 h before the F_2 was pumped out. The third section consisted of the flow meter, metering valve and molecular sieve traps used to regulate and purify the Ar flow. The flow reactor was separated from the mechanical pump (1000 liter/min) by a throttling valve (large stopcock), which was partially closed for the high pressure experiments. The reactor was of the coaxial mixing type frequently used by this laboratory[19] for observation of emission excited by interaction with metastable atoms. The Kr and Xe metastable atoms were produced by adding small amounts of Kr or Xe to the argon flow as described in the previous section. The hollow cathode discharge consisted of two rolled tantalum foil electrodes (operated at 250 V, ~1 mA) located 5.5 cm upstream from the mixing zone.

The reagent gases were obtained from the same suppliers as those used for the rate constant determination experiments. However, a higher grade of Ar (purity ≥ 99.995%) was used to reduce the problem of interfering impurity emissions, $N_2(C-B)$ and $OH(A-X)$. The OH^* emission from water impurity was still troublesome for Ar^*, and to a lesser extent for Kr^*, even with the precautions used here. The fluorine donors were diluted in Ar (mole fractions between 0.25 and 0.45) and stored in the passivated stainless reservoirs. Reagents were added coaxially to the flow of Kr^* or Xe^*; the resulting emission formed a diffuse flame some 10 cm in length. All observations were made with the entrance slit of the monochromator parallel to the gas flow and against the spectrosil quartz window, which was 2.5 cm below the origin of the reaction flame. Care was taken to ensure that all of the observations were made in the concentration range where the excimer emission intensity varied linearly with the flow (concentration) of the halogen donor. In this apparatus serious deviation from linearity occurred for reagent flow > 5 μmoles min^{-1} (6.3×10^{12} molecule cm^{-3}), see Fig. 1, because the high reagent concentrations significantly reduced the metastable atom concentration.

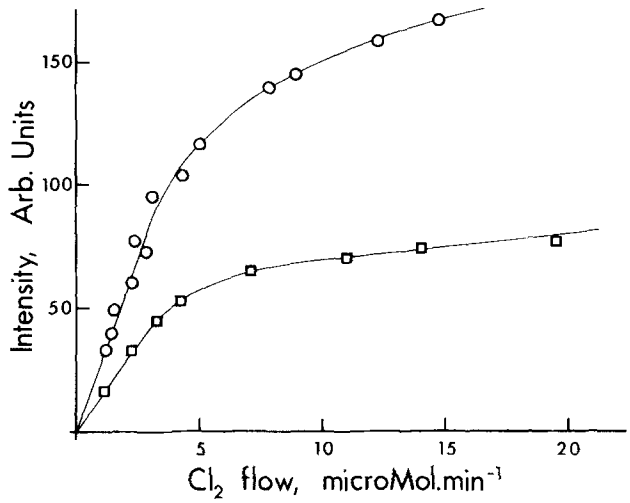


FIG. 1. Excimer emission intensity as a function of added Cl_2 flow: o -$XeCl^*$, monochromator slits = 150μ; □ -$KrCl^*$ monochromator slits = 120 μ. For the relative rate constant measurements, reagent flows never exceeded 3 μmoles min^{-1}. A flow of 5 μmoles min^{-1} corresponds to 6.3×10^{12} molecule cm^{-3}.

The emission was detected with a 0.75 m Jarrell-Ash Czerny-Turner scanning monochromator equipped with a cooled (0 °C) E.M.I. 9558 QB photomultiplier tube and SSR photon counter. The spectral response was calibrated from 300 to 850 nm using a standard quartz-iodine lamp (Electro Optics Associates) and by the "molecular branching ratio" method[4,20] from 300 to 190 nm. The resulting calibration curve was fitted piecewise to five (3 and 4 degree) polynomials using a nonlinear least square computer program. The calibration was checked and confirmed in the 190-300 nm region by using a standard deuterium lamp (Optronic Labs). The counter and monochromator were interfaced at a PDP/8e computer that controlled the scanning and data acquisition. The data were stored on magnetic tapes; these data were corrected for spectral response and reagent concentration and then integrated by simple addition of the counts in each wavelength increment. Normally the scan was taken with increments of 0.10 nm for low pressure excimer emissions, whereas high pressure spectra were taken with 0.05 nm increments. In order to test the reproducibility of our results, five independent experiments with constant $[Cl_2]$ were performed for $Xe(^3P_2)+Cl_2$ and the total integrated intensity agreed to within 5%. The reproducibility of experiments involving determination of areas of atomic lines was less, ~15%; this seemed to be related to wear in the monochromator screw drive mechanism.

Atomic absorption spectroscopy was used to obtain relative metastable atom concentrations for the quenching rate constant measurements. The relationship between absorption and concentration was carefully examined for $Ar(^3P_2)$ for $(I_0-I)/I_0 \leq 0.40$; the absorption of several atomic lines from the Pen-Ray lamp could be described[16] by a modified Beer-Lambert law for absorptions of up to 30%:

$$I = I_0 \exp[-a(k_0 l)^\gamma]; \quad \gamma = 0.95 \ , \qquad (1)$$

$$k_0 l = (2/\Delta\nu_D)(\ln 2/\pi)^{1/2}\,(\pi e^2/mc)\,[M^*]fl.$$

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TABLE I. Metastable atom concentrations from atomic absorption.

M	Transition	λ(nm)	f	A	$k_0 l$[a]	$[M^*]$[b]; 10^{10} atoms/cm^3
Ar[c]	$^3P_2$–$4p[\frac{5}{2}]_3$	811.5	0.380[f]	0.168	0.802	5.1
	$^3P_2$–$4p[\frac{3}{2}]_2$	763.5	0.223[f]	0.085	0.354	4.1
Kr	$^3P_2$–$5p[\frac{5}{2}]_2$	760.2	0.2575[d]	0.063	0.250	1.7
	$^3P_2$–$5p[\frac{5}{2}]_1$	810.4	0.1026[d]	0.028	0.106	1.7
	$^3P_2$–$5p[\frac{5}{2}]_3$	811.3	0.4839[d]	0.126	0.550	1.9
Ar	$^3P_2$–$4p[\frac{5}{2}]_3$	811.5	0.380[f]	0.193	0.965	6.1
	$^3P_2$–$4p[\frac{3}{2}]_2$	801.5	0.105[f]	0.036	0.140	3.3
	$^3P_2$–$4p[\frac{5}{2}]_2$	763.5	0.223[f]	0.100	0.420	4.8
Xe	$^3P_2$–$6p[\frac{5}{2}]_2$	881.9	0.4897[e]	0.118	0.510	1.3
	$^3P_2$–$6p[\frac{3}{2}]_2$	823.2	0.234[e]	0.052	0.203	1.1
Ar	$^3P_2$–$4p[\frac{5}{2}]_3$	811.5	0.380[f]	0.187	0.922	5.9
	$^3P_2$–$4p[\frac{5}{2}]_2$	801.5	0.105[f]	0.049	0.190	4.4
	$^3P_2$–$4p[\frac{5}{2}]_2$	763.5	0.223[f]	0.105	0.460	5.3
Xe	$^3P_2$–$6p[\frac{5}{2}]_2$	881.9	0.4897[e]	0.120	0.528	1.3
	$^3P_2$–$6p[\frac{3}{2}]_2$	823.2	0.234[e]	0.062	0.248	1.4

[a] $k_0 l$ was obtained from a prepared plot of A_α vs $\Sigma_n(-1)^n(1/n!) \times (k_0 l)^n/(1+n\alpha^2)^{1/2}$ for $\alpha = 3.5$.

[b] $[M^*] = 7.15(T/M)^{1/2}(k_0 l/f\lambda) \times 10^5$, λ in Å.

[c] The $^3P_2$–$4p[\frac{5}{2}]_2$ absorption was too small to be reliably measured for the first set of data.

[d] Calculated from transition probabilities of Ref. 22 using the relationship $f_{ik} = (A_{ki})\ 1.499\ \lambda^2\ (g_k/g_i)10^{-16}$, λ in Å.

[e] Calculated from transition probabilities of Ref. 23.

[f] See Ref. 16 for Ar f values.

The factor of 0.95 was determined experimentally from the slope of a $-\ln\ln(I_0/I)$ vs $\ln(k_0 l)$ plot, where the $k_0 l$ values were calculated using the more accurate and general expression (though more cumbersome) given by Mitchell and Zemansky,[21a]

$$A = \frac{I_0 - I}{I_0} = \sum_{n=1}^{\infty} (-1)^n \frac{(k_0 l)^n}{n!\,(1+n\alpha^2)^{1/2}}\ , \quad n = 1, 2, 3\ldots \tag{2}$$

The parameter α, the ratio of the widths of the emission and absorption lines, was determined[16] to be 3.5. For weak absorptions ($A < 0.2$), Eq. (2) is rather insensitive to the values of α between 3.0 and 4.0. Since the experimental conditions are similar for $Kr(^3P_2)$ and $Xe(^3P_2)$, except that the absorption is smaller than for $Ar(^3P_2)$, and since the source lamps are of the same type, Eq. (1) should adequately describe the absorption by $Kr(^3P_2)$ and $Xe(^3P_2)$ for the purpose of obtaining relative concentrations.

In addition to needing relative concentrations for rate constant measurements, the ratio of $Kr(^3P_2)$ to $Ar(^3P_2)$ and $Xe(^3P_2)$ to $Ar(^3P_2)$ are required for determining the branching ratios (see Sec. IIIC). One approach to measuring this ratio is to use atomic absorption spectroscopy and Eq. (1) to obtain absolute concentrations. These absorption measurements were done in the emission apparatus rather than in the rate constant apparatus, under conditions of constant pressure and geometry. The absorption for $Ar(^3P_2)$ first was measured; then either Kr or Xe added to the argon flow and the absorption for $Kr(^3P_2)$ or $Xe(^3P_2)$ was measured. In order to improve the reliability of the measurements, several atomic lines were used for each metastable atom. Representative results from several experiments are summarized in Table I. The variation in the $Ar(^3P_2)$ and $Kr(^3P_2)$ concentrations deduced from different lines reflects uncertainties in the atomic transition probabilities. However, the ratio of the mean value of $[Kr(^3P_2)]/[Ar(^3P_2)]$ is fully consistent with that expected from the $Ar(^3P_2) + Kr$ transfer reaction[24] followed by radiative branching of the $Kr^*(5p)$ states to the Kr(5s, $^3P_2$) and Kr(5s, $^3P_1$) levels.

The determination of the $Xe(^3P_2)$ concentration is more difficult because the values of the Xe(6s, $^3P_2$)–$Xe^*(6p)$ transition probabilities are not reliable. Allen *et al.* [25(a)] measured the lifetimes of the $Xe(6p[\frac{1}{2}]_0)$ and $Xe(6p[\frac{3}{2}]_2)$ states as (40 ± 12) and (33 ± 20) nsec, respectively, whereas Verolainen and Osherovich[25(b)] obtained (30 ± 4) and (48 ± 6) nsec for the same states. Calculated values[23] are ~28 and 29 nsec. The same calculations give f values for the 823.2 and 881.9 nm Xe lines of 0.23 and 0.49, which are in good agreement with the experimental results of Malakov,[25(c)] 0.29 and 0.5. This agreement is not very convincing because the calculated transition probabilities[23] are ~five times larger than the experimental values of Miller *et al.*[25(d)] for the Xe(6p′), Xe(7p), Xe(8p), Xe(7d), and Xe(8d) transitions terminating on the Xe(6s) and Xe(6s′) states. If the calculated values of Chen and Gastang are used, the concentrations shown in Table I are obtained. According to these results, the concentrations of $Kr(^3P_2)$ and $Xe(^3P_2)$ are approximately the same. For reasons that are discussed in Sec. IIIC, this ratio must be incorrect, and in fact the concentration of $Xe(^3P_2)$ is nearly the same as the concentration of $Ar(^3P_2)$. One possible reason for the low estimate for $[Xe(^3P_2)]$ from the absorption measurements is that the transition probabilities used in Table I are incorrect. Another possibility is that the large number of Xe isotopes with the resulting hyperfine effects on line widths complicate the interpretation of the observed absorption measurements.[21b]

III. RESULTS

A. Total quenching rate constants

When a reagent [Q] is present, the concentration of the metastable atoms $[M^*]$ at a position z in the reactor can be expressed by[16,26]

$$\ln\frac{[M^*]}{[M^*]_0} = -\{D_0/\Lambda^2 p + k_1 p + k_2 p^2 + k_Q[Q]\}\frac{z}{v}\ . \tag{3}$$

The first three terms arise from diffusion, two body, and three body processes in argon; the fourth term is the quenching by the added reagent Q which is present in sufficiently high concentration that pseudo-first-order kinetics are appropriate. Division of the flow distance z by the flow velocity v gives the reaction time. To obtain k_Q, Eqs. (1) and (2) are combined.

$$(1/\gamma)\ln\ln(I_0/I) \propto \ln[M^*]. \tag{4}$$

Therefore, for a fixed [Q] plots of $(1/\gamma)\ln[\ln(I_0/I)]$ vs z are linear. Figure 2 shows these decay curves with distance converted to time for the reaction of $Kr(^3P_2)$ with F_2. In obtaining the pseudo-first-order rate constants, the correction factor of 1.15, arising from converting distance to time, is included.[16] Plots of the pseudo-first-order rate constants K vs [Q] give straight lines with slopes k_Q. Figure 3 shows plots of

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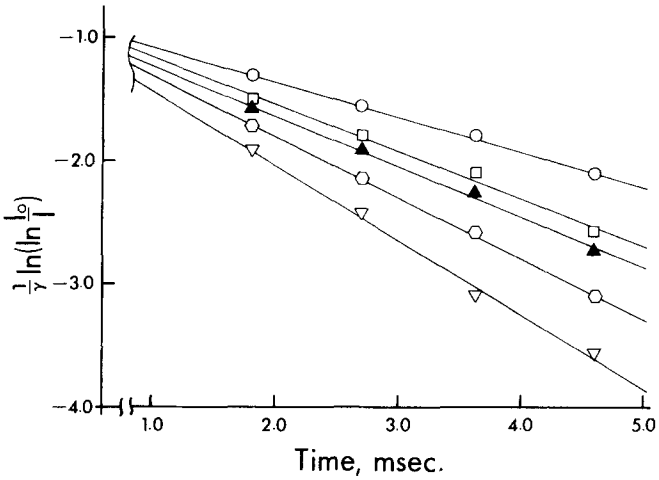


FIG. 2. "Decay" plots of $Kr(^3P_2)$ for various F_2 concentrations (molecules cm^{-3}) o -0.98×10^{11}; □ -2.17×10^{11}; ▲ -2.90×10^{11}; o -4.23×10^{11}; ▽ -5.98×10^{11}. Total pressure was 0.9 torr.

K vs [Q] for the data in Fig. 2, and for the reaction of $Kr(^3P_2)$ with NF_3 and $Xe(^3P_2)$ with CF_3OF. The intercepts in these plots are a result of the decay processes in pure argon. Least squares fits were used to obtain pseudo-first-order rate constants from the decay plots and quenching rate constants from the concentration plots. Typical standard deviations of k_Q were ± 5%.

All of the total quenching rate constants were obtained in a similar way and a listing is given in Table II. The majority of these values were calculated from one experiment at ~0.8 torr total pressure; however, tests for reproducibility showed that rate constants agreed well within < ± 10% for the same reagent mixture. The long term stability of the apparatus was tested by measuring the $Xe(^3P_2) + Br_2$ rate constant (by different operators) after an interval of 8 months; values of 60 and 63×10^{-11} cm^3 molecule^{-1} sec^{-1} were obtained. After a large number of rate constants were obtained, certain patterns were evident which suggested that some reagents needed reinvestigation, and Table I includes some reagents previously[16,17] studied. One problem in particular was that oil manometers were used to monitor flows in some of the previous work,[16] and this resulted in low rate constants for the reagents which dissolved in the oil. The Cl_2 reaction is one example, and the improved $Ar(^3P_2)$ rate constant for Cl_2 (71 vs 47×10^{-11} cm^3 molecule^{-1} sec^{-1}) is listed in Table II. Since the Cl_2 reactions with $Kr(^3P_2)$ and $Xe(^3P_2)$ were selected as the reference rate constants, each was measured with two mixtures; the reproducibility was ± 10%. Thus, we have confidence in the Cl_2 results of Table II. Since the F_2 reactions may be of importance for laser applications, measurements of the rate constants for $Kr(^3P_2)$ were done with two mixtures because the mixture composition is the most likely source of systematic error in the measurement of [Q]. The first mixture was prepared in our labora-

TABLE II. Quenching rate constants (10^{-11} cm^3 molecule^{-1} sec^{-1}).

	Ar $^3P_2$		Kr $^3P_2$		Xe $^3P_2$	
Reagent	k_Q	σ(Å^2)[a]	k_Q	σ(Å^2)[a]	k_Q	σ(Å^2)[a]
Cl_2	71[b]	142	73	179	72	193
F_2	75	132	72	146	75	161
Br_2	65	147			60	202
ICl	61	138			50	171
OF_2	57	107	53	121	57	139
CF_3OF	43	91	42	114	47	143
SF_6	16[b]	36			28[b]	94
SeF_6	71	166			65	246
TeF_6	58	135			63	230
SO_2F_2	42	89				
NF_3	14	28	8.9	22	9[c]	23
N_2F_4	31	65	33	90		
NOF	36[d]	68	47	102	39[d]	93
NOCl	48	95			51	135
HCl	37	65			56	119
HBr	52[b]	106			61	173
HI	75	166				
CF_3I					52	184
BrCN	46	100				
PCl_3	53	116				

[a]These cross sections were obtained from $k_Q/\langle v\rangle$, where $\langle v\rangle$ is the thermal average velocity of the collision pair.

[b]Taken from Refs. 16 and 17; see text for comments on Cl_2 and HBr.

[c]The low value for $Xe(^3P_2)$ was confirmed by experiments with a separate mixture.

[d]The rate constants for NOF with $Ar(^3P_2)$ and $Xe(^3P_2)$ may be somewhat low because the NOF mixture was stored in a glass reservoir. The rate constant with $Kr(^3P_2)$ was measured using the ss reservoir. Based upon the general trends in this table, the rate constants for $Ar(^3P_2)$ and $Xe(^3P_2)$ would not be expected to be larger than 50×10^{-11} cm^3 molecule^{-1} sec^{-1}.

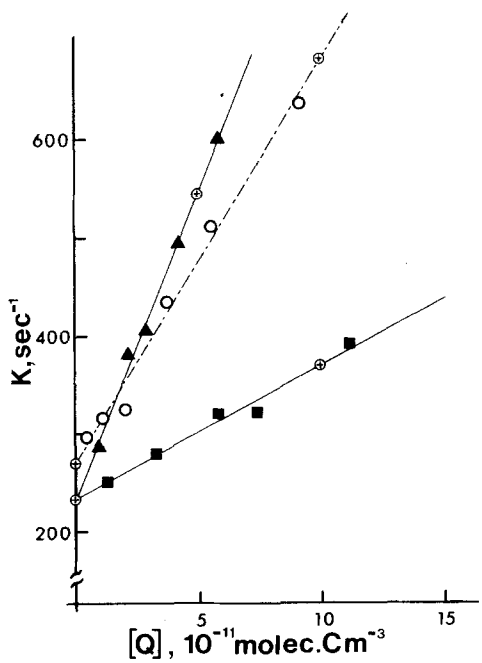


FIG. 3. Pseudo-first-order rate constants vs [Q]. Triangles, $Kr(^3P_2) + F_2$; squares, $Kr(^3P_2) + NF_3$; circles, $Xe(^3P_2) + CF_3OF$. The lines are least squares fits to the points. Open circles with a central + are calculated points from the least squares fit.

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tory; the second mixture was the commercially prepared mixture. The rate constant values for $Kr(^3P_2) + F_2$ from the two mixtures were 61 and 72×10^{-11} cm^3 molecule^{-1} sec^{-1}, respectively. The previously published value for HBr with $Ar(^3P_2)$ seems to be a case where mixture preparation led to error. Reinvestigation gave 52×10^{-11}, rather than the previous value of 77×10^{-11} cm^3 molecule^{-1} sec^{-1}. In the current work, care was taken to prepare HBr free from H_2 or Br_2 contamination, which is quite serious in samples from commercial lecture bottles.

With the exception of reagents with slow quenching rates, the rate constant values for a given reagent tend to be similar for all three metastable atoms. When transcribed into thermal quenching cross sections, a virtually constant ratio is obtained for a given reagent with the three metastable atoms since the mean speed is reduced for the heavier rare gas atoms.

B. Relative KrF and XeF excimer emission intensities

Collisional quenching of rare gas metastable atoms by molecules gives many exit channels, i.e., Penning ionization, associative ionization, excitation transfer, dissociation or dissociative excitation, and most notably chemical reaction with the production of $MX^* + R$:

$$\begin{aligned} M^* + RX &\xleftarrow{k_{PI}} M + RX^+ + e^- \ , \\ &\xrightarrow{k_{AI}} (MRX)^+ + e^- \ , \\ &\xrightarrow{k_{ET}} M + RX^* \ , \\ &\xrightarrow{k_D} M + R + X \ , \\ &\xrightarrow{k_{DE}} M + R^* + X \text{ (or, } M + R + X^*) \ , \\ &\xrightarrow{k_{MX}} MX^* + R \xrightarrow{\tau^{-1}} M + X + R + h\nu \ . \end{aligned} \tag{5}$$

The "Branching Ratios" are defined as

$$BR_i \equiv k_i \Big/ \sum_j k_j = \sigma_i \Big/ \sum_j \sigma_j \ , \tag{6}$$

where the k_i's and σ_i's are the rate constants (or cross sections) for the production of the given product, and $\sum_j k_j$ and $\sum_j \sigma_j$ are the total quenching rate constant k_Q and cross section σ_Q of Table II. For Xe^* and Kr^* with the fluorine donors, ionization is usually prohibited by energy conservation, but such processes are included here for generality. The main interest of the current work is the branching ratio for the formation of the rare-gas fluoride excimer k_{MX}/k_Q. Instead of attempting direct measurement of k_{MX}, a comparison technique was developed.

Provided that there is no quenching or nonradiative decay (such as predissociation[4]) of MX^*, the rate of photon emission will equal the rate of formation of MX^*,

$$I_{MX}^{RX} = k_{MX}[M^*][RX] \ . \tag{7}$$

If k_{MX} is assigned for a reference reaction, other MX formation rate constants can be obtained by comparing the total emission intensities:

$$\frac{I_{MX}^{R'X}}{I_{MX}^{RX}} = \frac{k'_{MX}[M^*][R'X]}{k_{MX}[M^*][RX]} \ . \tag{8}$$

If experiments are done for the same $[M^*]$, we obtain

$$k'_{MX} = k_{MX}\frac{I_{MX}^{R'X}/[R'X]}{I_{MX}^{RX}/[RX]} \ ; \quad \sigma'_{MX} = \left(\frac{\mu'}{\mu}\right)^{1/2}\sigma_{MX}\frac{I_{MX}^{R'X}/[R'X]}{I_{MX}^{RX}/[RX]} \ , \tag{9}$$

where μ and μ' are the reduced masses for the collision pairs $M^* + RX$ and $M^* + R'X$, respectively. In actual practice the measurement of absolute emission intensities is not necessary. All that is required is that the same fraction of the total emission be measured for each reaction. Of course, allowance for the variation of the detection efficiency vs wavelength must be made for each emission measurement.

Molecular chlorine was selected as the reference RX because the absolute values for k_{XeCl} and k_{KrCl} (or σ_{XeCl} and σ_{KrCl}) could be estimated, as will be shown in Sec. IIIC. A second reason for selecting Cl_2 was that it can be stored and metered in a reliable way, thus making the determination of concentrations more straightforward. The XeCl* and KrCl* emission spectra are shown in Figs. 4–6. The band system extending from the strongest maximum to shorter λ is called the primary emission; the weaker system to the red of the maximum will be called the secondary emission. The band on top of the KrCl secondary emission is the Cl_2^* 256 nm system, which also is produced from $Ar^* + Cl_2$ (Fig. 4) and very weakly from $Xe^* + Cl_2$. The Cl_2^* emission, which corresponds to 10% of the total KrCl* emission intensity, was subtracted when comparing the KrCl* emission intensity with that of KrF* from RF

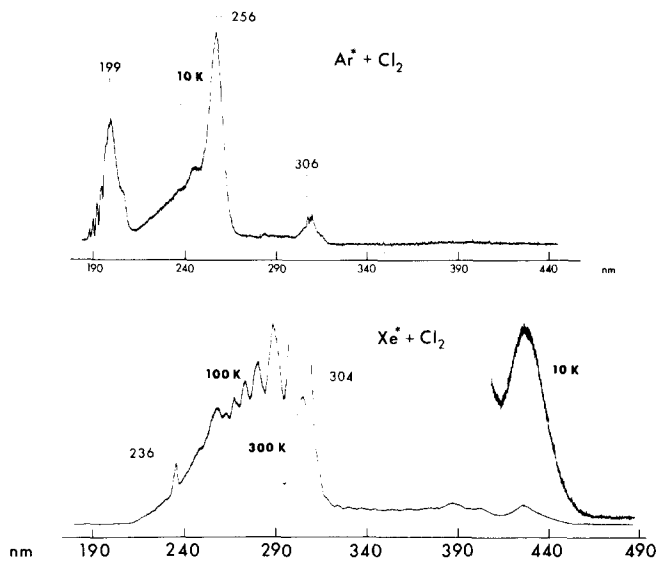


FIG. 4. Emission spectra from $Ar(^3P_{0,2}) + Cl_2$ (upper) and $Xe(^3P_2) + Cl_2$ (lower) at 1.5 torr; the different scales of the rate meter should be noted. For both cases the Cl_2 flow was $\sim 4.0 \times 10^{-6}$ mole min^{-1}. The main XeCl band begins at 304 nm and extends to shorter λ; a weaker system (secondary emission) extends to the red. The sharp 236 nm feature is emission from the excited $XeCl\ ^2\Pi_{1/2}$ excimer state. The 256 nm emission in both spectra (but stronger in $Ar + Cl_2$) arise from molecular Cl_2^*. An additional weaker Cl_2^* emission, superimposed with OH emission, occurs at 306 nm from $Ar^* + Cl_2$. The 199 nm emission from $Ar^* + Cl_2$ is the secondary band of ArCl*. Both the broad strong oscillations and the weak sharp oscillations (on the 100 K scale) of the XeCl* emission are real.

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reactions. One of the less desirable aspects of using KrCl* as the reference is that the emission extends to short wavelengths, where the monochromator calibration is less reliable. We estimate that ~10% of the KrCl* emission lies below the λ range observable with the Jarrell Ash monochromator; this correction balances that for the Cl_2^* 256 nm emission.

Based upon the constant ratio of the XeF* and NO(A–X) emission intensities from the Xe($^3P_2$) + F–NO reaction over the 1–40 torr pressure range (see Fig. 7), we concluded that argon pressures of up to 50 torr do not quench the XeF* excimer. However, the appearance of the spectra changes markedly with pressure above 3 torr as demonstrated in Figs. 5–7. These pressure effects, which have been discussed previously for ArCl[4], can be explained in part by vibrational relaxation of the excimer molecules initially formed in very high vibrational levels. However, since the ArCl* secondary emission also was reduced relative to the primary band with increasing pressure, collisional transfer between two ArCl* electronic states ($^2\Sigma^+$ and $^2\Pi_{3/2}$) also was suggested.[4] The secondary emission is very weak for XeF* and KrF*; however, for the chlorides the reduction in the secondary emission intensity is clearly observable in Figs. 5 and 6, and both electronic and vibrational relaxation may be occurring. The presence of a third excimer state ($^2\Pi_{1/2}$) can be noted by the small sharp band appearing in the spectra of Figs. 5–8. This excimer state (which was erroneously assigned to the molecular halogens in Ref. 3) and the

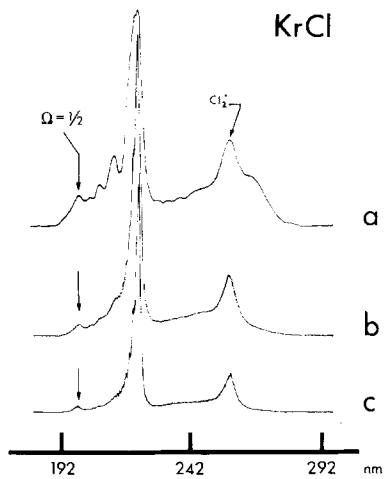


FIG. 6. Emission spectra of KrCl* at different total pressures. (a) 1.8 torr, (b) 15 torr, (c) 28 torr. For all cases the slits were 80μ and the scan speed was 12.5 nm/min. Some emission extends below the short wavelength limit of the monochromator for a. The Cl_2^* 256 nm band is superimposed on the KrCl* secondary emission. The sharp band at 198 nm corresponds to emission from the KrCl($^2\Pi_{1/2}$) excimer state. The main emission is *narrowed* and the secondary emission is *quenched* with increasing pressure; high resolution spectra show that the apparent structure at high pressure is not discrete bands.

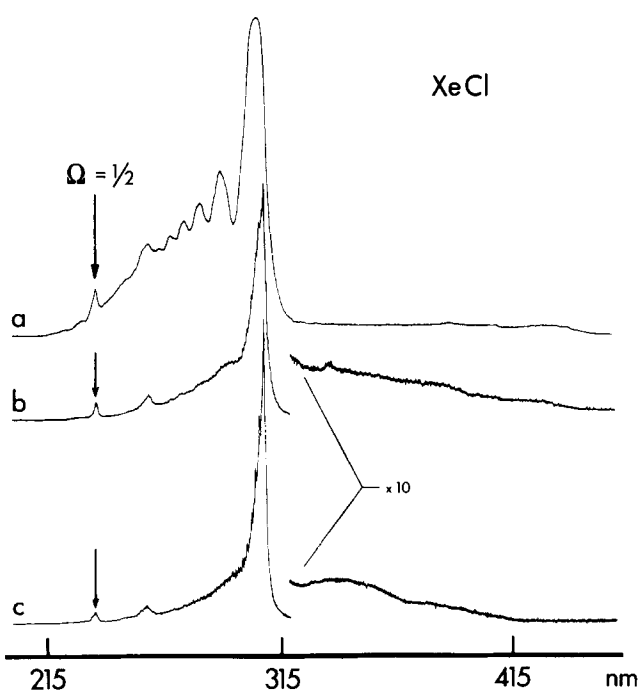


FIG. 5. Emission spectra of XeCl* at various pressure. (a) 1.7 torr, (b) 6 torr, (c) 26 torr. For all cases the slits were 55 μ and scan speed was 12.5 nm min^{-1}. At the highest pressure discrete bands from individual v' levels can be identified at λ = 307.7; 305.7; 304.0; 302.3; 300.7; 298.9, which correspond to 0–0, 1–0, 2–0, 3–0, 4–0, and 5–0 transitions.[8] The XeCl($^2\Pi_{1/2}$) emission is identified. The unlabeled band at 256 nm is Cl_2^* emission.

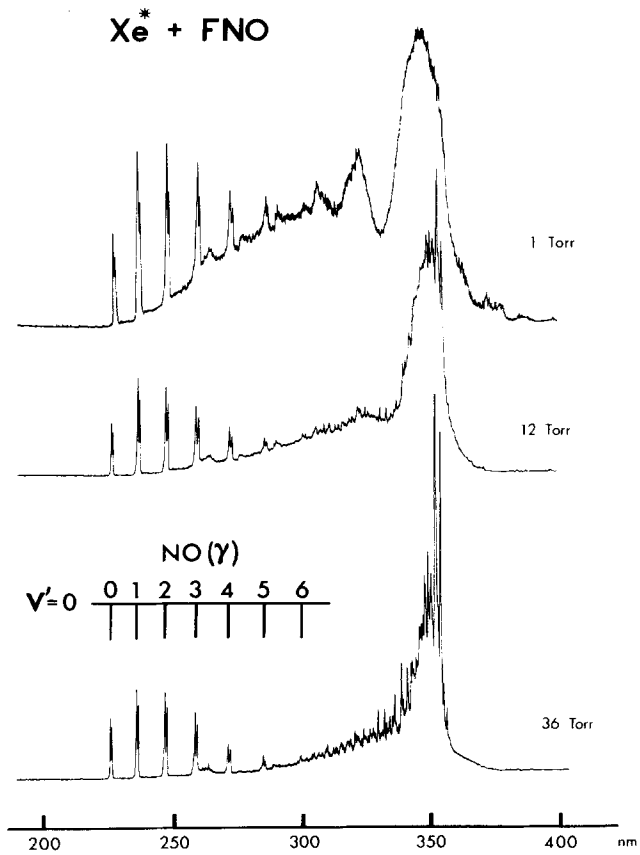


FIG. 7. Emission spectra from Xe($^3P_2$) +ONF at various pressures. Notice the extensive relaxation of the main band over the pressure range. The emission from the XeF($^2\Pi_{1/2}$) excimer state at 265 nm is still observable at 36 torr.

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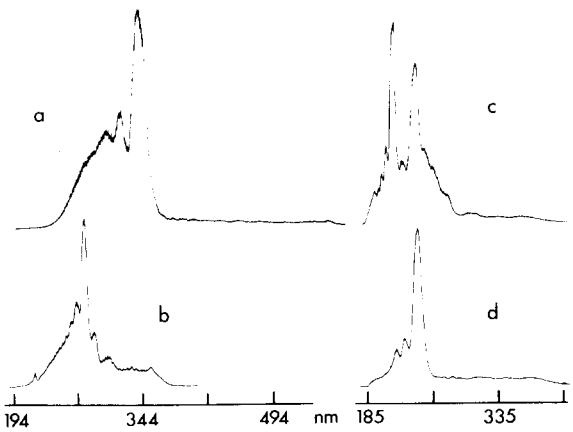


FIG. 8. Emission spectra from $Xe(^3P_2) + N_2F_4$ (a) and $Xe(^3P_2) + BrF_5$ (b) at 2 torr; the latter gives only XeBr*. Also shown are the emission spectra from the reaction of $Kr(^3P_2) + ClF_3$(c) and $Kr(^3P_2) + F_2$ (d) at 1.8 torr. Emission from both KrCl* and KrF* are clearly evident in (c). For all cases, the slits were 120 μ. The fast oscillations on the blue side of the main peaks of (a) are real; less pronounced oscillations also are evident in some of the other spectra.

changes of the spectra with pressure will be considered further in the discussion section. As a matter of convenience all of the intensity ratios were measured at 1–2 torr, and the integrated intensities include the full wavelength range of the excimer emission.

Some representative XeF* and KrF* spectra are shown in Figs. 7–9. Table III lists the relative formation rate constants (or relative emission intensities for equal concentrations of RX) for F donors grouped according to the location in the Periodic Table of the element bound to F. The relative emission intensities include the main band, the secondary band, and the third band ($^2\Pi_{1/2} - {}^2\Sigma^+$) contribution; the latter actually makes a negligible contribution for all reagents listed in Table III except for XeF_2. Also listed in Table III are the bond energies of the F–R donors and λ_{min}^{obs}, the observed short wavelength limit (at low pressure) for onset of the XeF* emission. The short wavelength limits of the KrF* emission are not given because the declining response of our detection system makes these impossible to observe. The relative intensities listed in Table III are the average of at least two scans performed during the same experiment, and in several cases (Groups VII and V) the average value includes experiments done weeks and even months apart. Immediately before and/or after each M* + RX experiment, a M* + Cl_2 experiment was run. For the same gas mixture, the reproducibility of the relative rate constants was ± 10%. In a given experiment, the main source of error was the instability of the discharge and the subsequent variation in [M*] over the 3–4 h period required to obtain scans for Cl_2 and RF. For the highly reactive fluoride donors, sample preparation and storage are the greatest source of error. The spectral response curve for the XeCl* and XeF* regions is thought to be highly reliable; however, the response declines rapidly at short wavelength and the comparison between KrCl* and KrF* could be subject to some systematic error. The relative KrF* intensities should be quite reliable.

Except for NOF, which gave NO(A and B) no other emissions were observed over the 800–200 nm region from RF reactions. Although all R–F molecules quench metastable atoms very readily, dissociative excitation with fluorine containing molecules is relatively rare for $Kr(^3P_2)$ and $Xe(^3P_2)$ reactions. Dissociative excitation also was found to be an unimportant exit channel for quenching of $Ar(^3P_{0,2})$ by chlorine containing molecules.[3]

C. Absolute determination of k_{MCL}

In order to put the relative rate constants of Table III on an absolute basis, the rate constant for formation of MCl* from the reaction of Cl_2 with $Xe(^3P_2)$ and $Kr(^3P_2)$ must be assigned. Since the absolute rate constants for the formation of excited Kr states by the reaction of $Ar(^3P_2)$ with Kr are known,[24] the Kr reaction was used to assign the MCl* formation rate constants in the following way. A reagent mixture of Cl_2 and Kr (normally 50-50) was reacted with $Ar(^3P_{2,0})$, and the relative emission intensities of the Cl_2^* 256 nm continuum and the 760.2 nm Kr* line were compared. Using the reported value[24] for the rate constant for formation of the $Kr^*[5p(\frac{3}{2})_2]$, which is the source of the 760.2 emission, the rate constant for formation of the Cl_2^* 256 nm continuum can be calculated. The average of 35 determinations with three mixtures gave $(2.0 \pm 0.2) \times 10^{-11}$ cm^3 molecule^{-1} sec^{-1}. At this point, the concentration of Ar* was measured using absorption spectroscopy. Then a flow of Kr was added (before the discharge) to the argon flow to form $Kr(^3P_2)$, and the concentration of $Kr(^3P_2)$ was measured using absorption spectroscopy, see Table I. Finally this Kr* concentration was reacted with the same flow of Cl_2 as used with $Ar(^3P_{0,2})$. The spectra using this procedure from Ar* + Cl_2 and Xe* + Cl_2 are shown in Fig. 4. To obtain

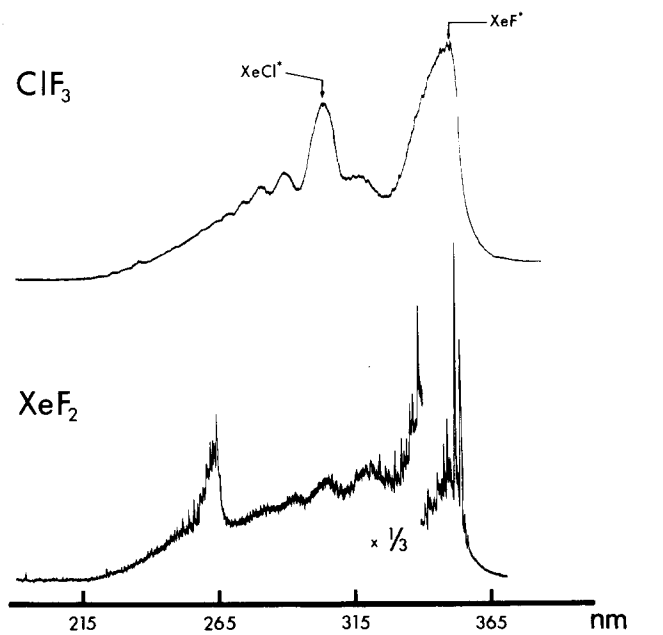


FIG. 9. Emission spectra form $Xe(^3P_2) + ClF_3$ (upper) and XeF_2 (lower) at 1.5 torr. Both XeF* and XeCl* are formed from ClF_3. Note that the XeF* emission from XeF_2 is more relaxed at 1.5 torr than the XeF* emission at 12 torr from Xe + FNO (Fig. 7).

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TABLE III. Relative rate constants for KrF* and XeF* excimer formation.

Group	Reagent	D_0^0 [a] kcal/mole	k_{Xe}	k_{Kr}	$\lambda_{min}^{obs} - \lambda_{min}^{cal}$, nm[b]
	Cl_2	57.9	100	100	214 – 214
VIII	XeF_2	61[c]	Strong[d]	Strong[d]	217–219
VII	F_2	38	100	88	190–186
	ClF	60	17(87)[f]	16(71)[f]	220–217
	ClF_3	54[e]	20(33)[f]	34(56)[f]	207–
	BrF_5		0(23[g])[f]	0(33[g])[f]	193–
	IF_5		0(~5[g])[f]		205–
VI	OF_2	40[h]	70	69	202–188
	CF_3OF	44[i]	25	30	205–194
	SF_6	$\gtrsim$ 78	~0	~0	
	SOF_2	~86[j]	~1	~1	308–271
	SO_2F_2	~84[j]	0	0	
	SF_4	~83[k]	0	0	
	SeF_6		0	0	
	TeF_6		0	0	
V	NF_3	57	14	7	245–212
	N_2F_4	71[l]	27	20	230–237
	ONF	55	40, (43)[m]	<5[n]	220–209
	PF_3		<1.0		
IV	CF_4	125	0	0	
	SiF_4	110	0	0	
III	BF_3	134	0	0	

[a]Unless indicated otherwise, the bond energies were taken from Ref. 27.
[b]λ_{min}^{obs} and λ_{min}^{cal} are the observed and calculated, $E(Xe, {}^3P_2) - D^0(R-F)$, short wavelength limit of the XeF* or XeCl* emission.
[c]Calculated from $\Delta H_f^0(XeF_2) = 65$ kcal mole^{-1}, assuming $D_0^0(XeF) = 4$ kcal mole^{-1}; see Ref. 18.
[d]Since the XeF_2 flows were not measured accurately, relative rate constants can not be assigned.
[e]Calculated from the observed $\lambda_{min}^{obs}(XeF)$; therefore this value is an upper limit.
[f]First number represents relative rate constant for XeF* formation, number in parenthesis represents total excimer emission. The approximate ratio of I_{XeF^*}/I_{XeCl^*} was 1.5 (see Fig. 8) for ClF_3 and 0.25 for ClF.
[g]Only XeBr*, KrBr,* or XeI* emission was observed.
[h]Average of the values reported in Ref. 29.
[i]Reference 28.
[j]Average bond energy calculated from heats of formation.
[k]See Ref. 31.
[l]Reference 30.
[m]Includes emission from NO(γ) bands, see Fig. 7.
[n]The NO(β) and NO(γ) bands consitute most of the emission; the listed value corresponds to the estimated KrF* emission intensity.

the k_{KrCl} rate constant only the ratio of the $Ar({}^3P_{0,2})$ and $Kr({}^3P_2)$ [or $Xe({}^3P_2)$] concentrations is needed, e.g.,

$$\frac{I_{Cl_2}^*}{I_{KrCl}^*} = \frac{k_{Cl_2}^{Ar^*}}{k_{KrCl}} \frac{[Cl_2]}{[Cl_2]} \frac{[Ar({}^3P_{0,2})]}{[Kr({}^3P_2)]} \quad . \tag{10}$$

Both $Ar({}^3P_2)$ and $Ar({}^3P_0)$, which are present in a ~6 : 1 ratio, are assumed to give Cl_2^*. Combining the relative intensities of the Cl_2^* 256 nm continuum [from $Ar({}^3P_{2,0})$] and the KrCl* emission with the measured Ar* and Kr* concentrations of Table I gave $k_{KrCl} = 74 \times 10^{-11}$ cm^3 molecule^{-1} sec^{-1}. If the rate constant for Cl_2^* formation by $Kr({}^3P_2)$ is included, a total quenching rate constant of 81×10^{-11} cm^3 molecule^{-1} sec^{-1} is obtained. Although this value is close to the measured total quenching rate constant, some caution should be exercised. The comparison of the Cl_2^* 256 nm continuum and the Kr* 760.2 nm line intensities from $Ar({}^3P_2)$ are subject to some uncertainty because the comparison uses the ends of our calibrated spectral response curve, which could be in error by ±20%. As discussed in Sec. IIB, we have reasonable confidence in the measured $[Kr({}^3P_2)]/[Ar({}^3P_2)]$ ratio; however, the uncertainty still could be ±30%. Within these experimental limitations, the branching ratios for formation of KrCl* and Cl_2^* are assigned as 0.90 and 0.10, respectively.

The determination of the $[Xe({}^3P_2)]/[Ar({}^3P_2)]$ ratio is more difficult because the interpretations of the absorption measurements, Table I, are less reliable. As an added complication, the experiment to be described below shows that $Xe({}^3P_2)$ is produced directly in the discharge, as well as by radiative cascade following the $Ar({}^3P_{0,2}) + Xe$ excitation transfer reaction.[32] In order to judge the ratio of $Xe({}^3P_2)$ and $Kr({}^3P_2)$ concentrations, an experiment was done in which the XeCl* and KrCl* and the XeF* and KrF* emission intensities were compared from $Xe({}^3P_2)$ and $Kr({}^3P_2)$ that were obtained from the same Ar* concentration and from the same Cl_2 and F_2 concentrations. The emission intensities of XeF* and XeCl* were *twofold* larger than that for KrCl* and KrF*. Since the branching ratios for the $Kr({}^3P_2)$ reactions already have been shown to be ~ unity, $[Xe({}^3P_2)]$ must be greater than $[Kr({}^3P_2)]$. Since the $Ar({}^3P_2)$ transfer reaction cannot yield $Xe({}^3P_2)$ with unit conversion efficiency,[32] $Xe({}^3P_2)$ must be directly produced in the discharge. This implication was confirmed by comparing the XeCl* emission intensities when $Xe({}^3P_2)$ was prepared by adding Xe to the argon flow before and after the discharge. Enough Xe was added to ensure that excitation transfer was completed before the Xe/Ar mixture reached reaction zone. The emission intensity for addition of Xe before the discharge was 2.4 times larger than for the addition after the discharge, which conclusively proves that direct production of $Xe({}^3P_2)$ occurs in the discharge. A similar experiment with Kr gave ≤15% increase in KrCl* emission intensity for addition of Kr to the argon flow before the discharge. We conclude that the higher XeCl* and XeF* emission intensities, relative to KrCl* and KrF*, can be attributed to the larger $Xe({}^3P_2)$ concentration and that the branching ratio for XeCl* and XeF* formation from Cl_2 and F_2 must be very close to unity.

In summary, branching ratios for formation of XeF* and KrF* from F_2 were assigned as unity for the following reasons: (i) Within experimental uncertainty the measured value of the KrF* *formation rate constant* (65×10^{-11} cm^3 molecule^{-1} sec^{-1}) equals the *total quenching rate constant*. (ii) The ratios of the relative rate constants for XeF* and XeCl* formation from F_2 and Cl_2 are the same as for the total quenching rate constants. Also, the absolute comparisons noted above strongly suggest (but do not prove) a unity branching ratio for both Cl_2 and F_2 reactions with $Xe({}^3P_2)$. (iii) No other emissions were observed from the F_2 reactions.

Based upon the assignment of unit branching ratios

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TABLE IV. Branching ratios[a] for KrF* and XeF* formation.

Reagent[c]	$Xe(^3P_2)$	$Kr(^3P_2)$
F_2	1.0	1.0
OF_2	0.92	1.06
CF_3OF	0.40	0.58
NF_3	1.16	0.57
N_2F_4	0.63[b]	0.50
NOF	0.77	<0.09

[a]The uncertainties are estimated as ±0.20.
[b]The k_q value was taken to be the mean of the rate constants for $Ar(^3P_2)$ and $Kr(^3P_2)$.
[c]Eximer formation from ClF has probably unit branching ratio. However, the main product is the chloride.

for F_2, the branching ratios for the better fluorine donors are listed in Table IV. The branching ratios with XeF_2 could not be assigned because of the difficulty of measuring the XeF_2 flow; however, on a qualitative basis the XeF* emissions were quite strong and the branching ratio must be large. Although ClF probably has a unit branching ratio for rare gas halide excimer formation, the favored exit channel is the chloride excimer rather than the fluoride excimer; therefore, ClF is not listed in Table IV. The extinction coefficients of the better fluorine donors of Table IV do not appear to be so large as to interfere with the XeF or KrF laser performance.

IV. DISCUSSION

A. Rare gas excimer states for krypton and xenon halides

The states of the excimers will be discussed by using the calculated[33] potentials of KrF as reference. As shown in Fig. 10, three states correlate with $Kr(^1S_0) + F(^2P_{3/2})$ and $F(^2P_{1/2})$ and three states correlate with $F^-(^1S_0) + Kr^+(^2P_{3/2})$ and $Kr^+(^2P_{1/2})$. The latter are essentially Coulombic states. A fourth bound state (the lowest bound Rydberg state) lies in the 9–10 eV range and correlates with $F(^2P_{3/2})$ and $Kr(^3P_2)$; this state does not play a role in the present discussion but is important in quenching of metastable rare gas atoms by atomic halogen.[32] According to Dunning and Hay's calculations, the lowest excimer state is the $^2\Sigma^+$ (or $\Omega=\frac{1}{2}$) state. However, the calculated separation between the $^2\Pi_{3/2}$ (or $\Omega=\frac{3}{2}$) and the $^2\Sigma^+$ states is very small for KrF* and there still may be some question about this ordering for other rare gas halides. The main rare gas halide excimer emission is usually assigned as $(^2\Sigma^+-{}^2\Sigma^+)$; however, for the low pressure spectra observed in our work this transition could be overlapped by weak emission from $^2\Pi_{3/2}-{}^2\Sigma^+$. The secondary emission (that portion of the emission extending to the red of the strongest band) is understood less well but probably involves emission from $^2\Pi_{3/2}$ to $^2\Pi_{1/2,3/2}$ with possibly some contribution from $^2\Sigma^+$ to $^2\Pi_{1/2,3/2}$. The secondary emission is quite pressure dependent, *vide infra*. Both the main and secondary emission generated from reactive quenching at low pressure exhibit some interference patterns[5] resulting from the nature of the bound–free transitions from a distribution of vibrational levels. As the pressure is increased, vibrational relaxation occurs and for sufficiently high pressure the emission is from a Boltzmann distribution of levels. Under such conditions the XeF* and XeCl* emissions are bound–bound and give the usual vibrational bands from which spectroscopic constants can be obtained.[8] For some of the rare gas halides, especially ArCl*, ArBr*, and ArI*, the bound Coulombic states are very likely predissociated by diabatic states correlating to rare gas + excited halogen states.[4] This is not of importance for Kr or Xe with F or Cl.

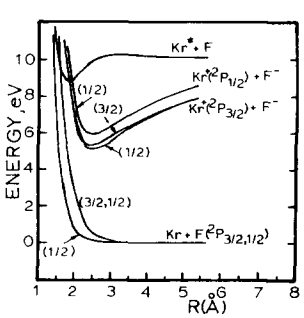


FIG. 10. Calculated KrF potentials. The results were taken from Ref. 33.

The main contribution of the present work to further elucidation of the excimer states is the identification of the $^2\Pi_{1/2}(\Omega=\frac{1}{2})-{}^2\Sigma^+$ emission. A definitive spectrum (see Fig. 9) was obtained from the reaction of $Xe(^3P_2) + XeF_2$; the general appearance of the 260 nm band is very similar to that of the main emission at 350 nm and the 260 nm band clearly can be assigned to the $^2\Pi_{1/2}-{}^2\Sigma^+$ transition, as previously suggested by Ewing and Brau.[6] After obtaining the 260 nm XeF* spectrum from XeF_2, it was possible to identify this transition, which normally is very weak, in spectra from many other halogen donors with low band energies. The results for Xe with all halogens and Kr with Cl and F are tabulated in Table V. As shown in Table V, this energy difference is nearly constant for the various halogens. Furthermore, the energy separation is nearly the same as the asymptotic separation, as might be expected since the bonding is dominated by Coulombic forces. The energy difference is actually slightly less than the asymptotic separation, so there apparently is more repulsion in the $^2\Sigma^+$ than in the $^2\Pi_{1/2}$ state, i.e., the well depth of $^2\Pi_{1/2}$ is larger. Since the $^2\Pi_{1/2}$ and $^2\Sigma^+$ states have nearly the same separation at R_e as in the asymptotic ion pair limit, the energy separation at

TABLE V. The $^2\Pi_{1/2}-{}^2\Sigma^+$ excimer transition.

Molecule	Wavelength (nm)[a]	Energy (eV)	Separation[b] from $^2\Sigma^+$ (eV)
XeF	264.2	4.702	1.16
XeCl	235.8	5.269	1.21
XeBr	221.3	5.615	1.20
XeI	203.0	6.123	1.21
KrF	220.5	5.636	0.65
KrCl	198.7	6.255	0.65

[a]For XeF, this wavelength was measured at the sharp band that resembles the 351.1 nm band of the main emission. For all others, the wavelength corresponds to the maximum of the observed sharp peak of the $^2\Pi_{1/2}-{}^2\Sigma^+$ transition.
[b]The separation of the $^2P_{1/2}$ and $^2P_{3/2}$ states of Xe^+ and Kr^+ are 1.307 and 0.665 eV, respectively. Except for XeI, the separation was calculated from the laser transitions and the wavelengths tabulated in column 2. For XeI, 253 nm was used as the head of the main emission.

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R_e between the $^2\Pi_{3/2}$ and $^2\Sigma^+_{1/2}$ states (which correlated to the same asymptotic limit) may be quite small as shown by the calculated results in Fig. 10.

The secondary emission, which is thought to represent termination on the $^2\Pi$, rather than the $^2\Sigma^+$ lower state, is very weak for XeF* and KrF*. Therefore, attention can be focused on just the variation of the main emission with pressure. As shown in Fig. 7, the XeF emission intensity shifts toward the long wavelength limit with increasing pressure and acquires structure characteristic of a bound–bound emission originating from the lower vibrational levels of the upper state. Since emission from high vibrational levels of XeF* is mainly bound–free, the spectrum from a broad distribution of levels consists of both bound–free and bound–bound contributions. Vibrational relaxation occurred without any significant change in the relative integrated intensity of the main XeF* and NO(A–X) emission. Since the NO(A) state is not quenched by argon, the data of Fig. 7 show that more than 40 torr of Ar is required for quenching of XeF*. The observation of extensive vibrational relaxation over the pressure range 1–10 torr implies that the radiative lifetime is ≥ 50 nsec, since the time between gas kinetic collisions is ~ 50 nsec at 5 torr. This estimate is a factor of 2 larger than the value obtained for XeBr* by Searles and Hart[10] from pulsed e-beam excitation and an order of magnitude larger than the calculated value for KrF* by Dunning and Hay.[33]

The secondary emission, which has greatest intensity for combinations of the *heavier* halogens and *lighter* rare gases, remains a perplexing problem. The secondary KrCl* emission, upon which the Cl_2^* 256 nm continuum is superimposed, is clearly evident in Fig. 6. As the pressure is increased, the secondary emission intensity declines and shifts to the blue and at the highest pressure the Cl_2^* band has become dominant. The decline of the secondary emission intensity relative to primary emission intensity with increasing pressure previously has been noted[4,5] for ArCl*, ArF*, and KrBr*. Also, the ratio of the main and secondary ArCl* emission intensities varied with donor.[4] Collisional transfer between the $^2\Sigma^+$ and $^2\Pi_{3/2}$ electronic states, as well as vibrational relaxation, were suggested for ArCl*. The present work provides no further insight but emphasizes that the secondary emission and its variation with pressure are not well understood. Since the Xe($^3P_2$) and Kr($^3P_2$) reactions with RF are not expected to be any more specific than with RCl for formation of given excimer states, the different primary and secondary intensity ratios for the fluorides and chlorides implies that the radiative branching ratios must change for the fluorides vs the chlorides. Thus the question becomes: Why are the radiative transitions terminating on the $^2\Pi_{1/2,3/2}$ states less favorable for the fluorides than for the chlorides?

Ewing and Brau[7] reported a broad smooth emission system near 450 nm for e-beam excitation of Xe and F_2 mixtures. The XeF* secondary emission resulting from direct reaction between Xe($^3P_2$) and RF at low pressure is observed as a very long, but flat, tail extending from 360 to 500 nm. This low pressure secondary emission does not appear to be the same as that observed by Ewing and Brau.

B. Reactive quenching of rare gas metastable atoms by RF and Cl_2

Efficient formation of XeF* and KrF* evidently only occurs for a limited choice of fluoride donors, e.g., fluorine and small molecules with the OF or NF bond. Detailed consideration (e.g., arguments of electron affinity, nature of the lowest unfilled orbital, stability of RF^-, etc.) of the reaction mechanism for reactive quenching with formation of KrF* or XeF* will be delayed until after the other halide donors have been studied. Only a general summary will be attempted with emphasis on the decline in the branching ratio as RF becomes larger. The general interaction mechanism with most RF molecules appears to be a charge transfer mechanism in which the entrance channel $V(M^*, RF)$ is crossed at relatively long range by an ion pair potential $V(M^+, RF^-)$. Efficient transfer to $V(M^+, RF^-)$ is followed by competition between formation of MF* + R and M + RF*. The RF* channel is observed as either molecular fluorescence, e.g., Cl_2^*, or as dissociation with excitation of R*. For most RF reagents dissociation of RF* must occur without excitation of R*; however, there are other systems (such as RCN)[34] that give a large amount of dissociative excitation. For diatomic halogens the reactions are expected to be direct with the MX* channel being dominant. This expectation is realized with Cl_2, F_2, and ClF since branching ratios are nearly unity, and high vibrational levels are formed as indicated by observation of emission from MX* levels that correspond to the thermochemical energy limit. However, for larger RF some evidence suggests that M^+RF^- may be longer lived. Excimer formation occurred from the *central* atom with BrF_5 and IF_5, and some time presumably is required for this rearrangement. Furthermore, the XeF($^2\Sigma$ or $^2\Pi_{3/2}$) product formed from XeF_2 was more vibrationally relaxed than from N_2F_4, which has about the same bond energy, suggesting a lifetime sufficiently long for some dissipation of energy into internal modes of the Xe^+–XeF_2^- complex. Thus, the reaction dynamics for metastable rare gas atoms with RF probably includes cases proceeding via long-lived complexes,[35] as well as those proceeding by direct reaction. If the charge–transfer complex has an increased lifetime with respect to entering the MF* channel, the RF* channel seems to dominate as the exit channel. The probability for the latter also may increase with larger R because of the greater number of excited RF* potentials. With this qualitative view, it is not surprising that the branching ratios for excimer formation only approach unity for small molecules with low bond energies. Based upon the formation of the Cl_2^* state giving the 260 nm continuum from the Ar($^3P_2$) and Kr($^3P_2$) + Cl_2 reactions, escape from the charge transfer potential $V(M^+, RF^-)$ to an excited RF potential can be quite rapid.

There is an alternative explanation to the reduced branching ratio for excimer formation with larger RF donors that does not depend upon formation of RF* states from the charge transfer intermediate. The efficiency of transfer from $V(M^*, RF)$ to $V(M^+, RF^-)$ could decline with increasing size of RF. Quenching then would oc-

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cur at shorter range by direct transfer from $V(M^*, RF)$ to $V(M, RF^*)$. Although we do not generally favor this explanation, it may be of importance for some reagents since all collision pairs may not have orientations that favor the transfer from $V(M^*, RF)$ to $V(M^+, RF^-)$.

Reactions known to occur via the direct long-range ionic curve crossing mechanism release a large fraction of the available energy to vibrational energy of the halide product.[36] If the MX* excimers are formed by this mechanism, the emission should extend to the thermochemical limit. As already noted, this certainly is true for $Xe(^3P_2) + Cl_2$, F_2, and ClF. Within the uncertainty of the bond energies and the assignment of λ_{min}^{obs} [care must be taken to ensure the total absence of $Ar(^3P_{0,2})$ since the ArF* secondary emission overlaps the tail of the main XeF* emission], the XeF* emission from the FNO, OF_2, and CF_3OF reactions also extend to the thermochemical limits. The situation for NF_3 and N_2F_4 is more complicated. The XeF* emission from N_2F_4 extends to *smaller* λ than was expected for a bond energy of 71 kcal mole^{-1} and these data, as well as the work of Golde,[5] suggest that $D^0(F-N_2F_3)$ is less than 71 kcal mole^{-1}. Based upon these observations, the $Xe(^3P_2) + N_2F_4$ reaction appears to be direct. Although $D^0(F-NF_2) \sim 57$ kcal mole^{-1} is generally accepted,[29] the XeF* emission clearly does not extend to the thermochemical limit. The NF_3 reaction seems unlikely to involve a long-lived complex if the N_2F_4 reaction is direct. One explanation could be that NF_2 is a stabilized radical and that the radical stabilization energy is not released[37] to the XeF* product. The $Xe^* + XeF_2$ reaction merits special consideration. The reaction obviously yields $XeF(^2\Sigma^+$ or $^2\Pi_{3/2})$ in high vibrational levels. In fact, the emission does extend to the thermochemical limit (the low ~4 kcal mole^{-1} value for $D^0(F-Xe)$[8b] gives a large $D^0(F-XeF)$ value of 60 kcal mole^{-1}).[38] The $Xe^* + XeF_2$ reaction may proceed by two different pathways. One may be the direct reactive quenching and the second may be excitation transfer followed by predissociation of XeF_2^*. Another possibility is that the Xe^+, XeF_2^- complex breaks up by two different pathways, which may correlate with the different Xe atoms in the complex.

The total reactive cross sections of the metastables with halogens (at 300 °K) can be compared with those of the alkali metals[39] (at ~1000 °K). The values for Cl_2 with K(Ar*), Rb(Kr*), and Cs(Xe*) are 154 (142), 190 (179), and 196 (193) Å^2, respectively. For Br_2 the comparison gives 151 (147), 197 (?), and 204 (202) Å^2. The similarity is quite remarkable. The variation of the cross sections with temperature is small[38] and will not significantly affect these comparisons. Based upon an optical model, the differential scattering data of $Ar(^3P_{0,2})$ with HBr have been analyzed[40] to obtain total inelastic cross sections at four energies in the range 60–160 meV. The thermally averaged value from that work gave 14.7 Å^2. Although our previously published value was about 20% too high, the cross section in Table II is still ~8 times larger than the results from the beam work.[40] The total quenching cross sections for all of the hydrogen halides exceed 100 Å^2, and the inelastic cross section deduced from the beam

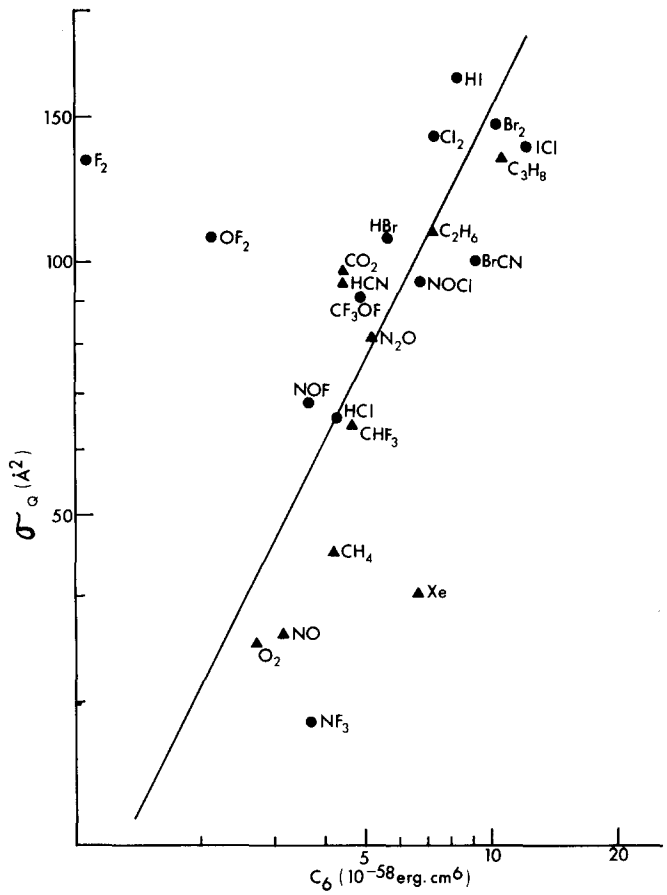


FIG. 11. Metastable $Ar(^3P_2)$ quenching cross sections vs the C_6 coefficient of the r^{-6} attractive intermolecular potential. For convenience of display the scales of the axis are not equal; the line still is of unit slope. The symbols o denote cross sections measured in this work; the ▲ points were previously reported[16] values.

measurements must be questioned.

In earlier work[16] a linear correlation of total quenching cross sections σ_Q vs the long range attractive van der Waals coefficient C_6 was found for a variety of reagents but that selection of reagents included only a few halogen-containing molecules. A similar plot is shown in Fig. 11 for the quenching of $Ar(^3P_2)$ by halogen-containing molecules. The C_6 values were calculated using the Slater–Kirkwood approximation. The polarizabilities used in the calculations were obtained from the literature[41,42] or were estimated using additive values given by Applequist, Carl, and Fung.[42] The effective number of electrons used in the Slater–Kirkwood approximation was taken to be the sum of the valence shell electrons of the elements making up the molecule in question. Contributions to C_6 from permanent dipoles were not included in the calculation. The calculated C_6 values will have an uncertainty of ~20%. Somewhat surprisingly the correlation plot for the halogen-containing molecules still is remarkably good. Since the ratios of cross sections for $Ar(^3P_2)$, $Kr(^3P_2)$, and $Xe(^3P_2)$ are nearly constant,[43] a similar correlation plot also holds for the quenching of $Kr(^3P_2)$ and $Xe(^3P_2)$. Notable exceptions to the correlation are F_2 and to a lesser extent OF_2. The large cross sections of F_2 and OF_2, in spite of their low polarizabilities, can be ex-

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plained in terms of direct reaction via the long range, curve-crossing mechanism between the entrance channel and charge–transfer potential. The more puzzling question is why the other halogens follow this C_6 correlation so closely, when we have just argued that quenching occurs by the charge-transfer, curve-crossing mechanism. A tentative explanation, pending detailed investigation, is that the inclusion of the long range attractive interactions into the curve-crossing model (which then evolves to an orbiting collision model)[39] will have effects which result in the correlation noted in Fig. 11.

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